EXERCISES

QUESTION 1:

Length and breadth of a rectangle are 5 and 7 respectively. Write a program to calculate the area and perimeter of the rectangle.

CODE:

```
#include<iostream>
using namespace std;
int main(){
	int l(5),b(7);
	int p=(l+b)*2;
	int area=l*b;
	cout<<"Length is: "<<l<<endl
		<<"Breadth is: "<<b<<endl;
	cout<<"Perimeter of Rectangle is: "<<p<<endl
		<<"Area of Rectangle is: "<<area;
}
```

OUTPUT:

```
C:\Users\Vishal\Desktop

Length is: 5
Breadth is: 7
Perimeter of Rectangle is: 24
Area of Rectangle is: 35
--------------------------------
Process exited after 0.07895 seconds with return value 0
Press any key to continue . . .
```

QUESTION 2:

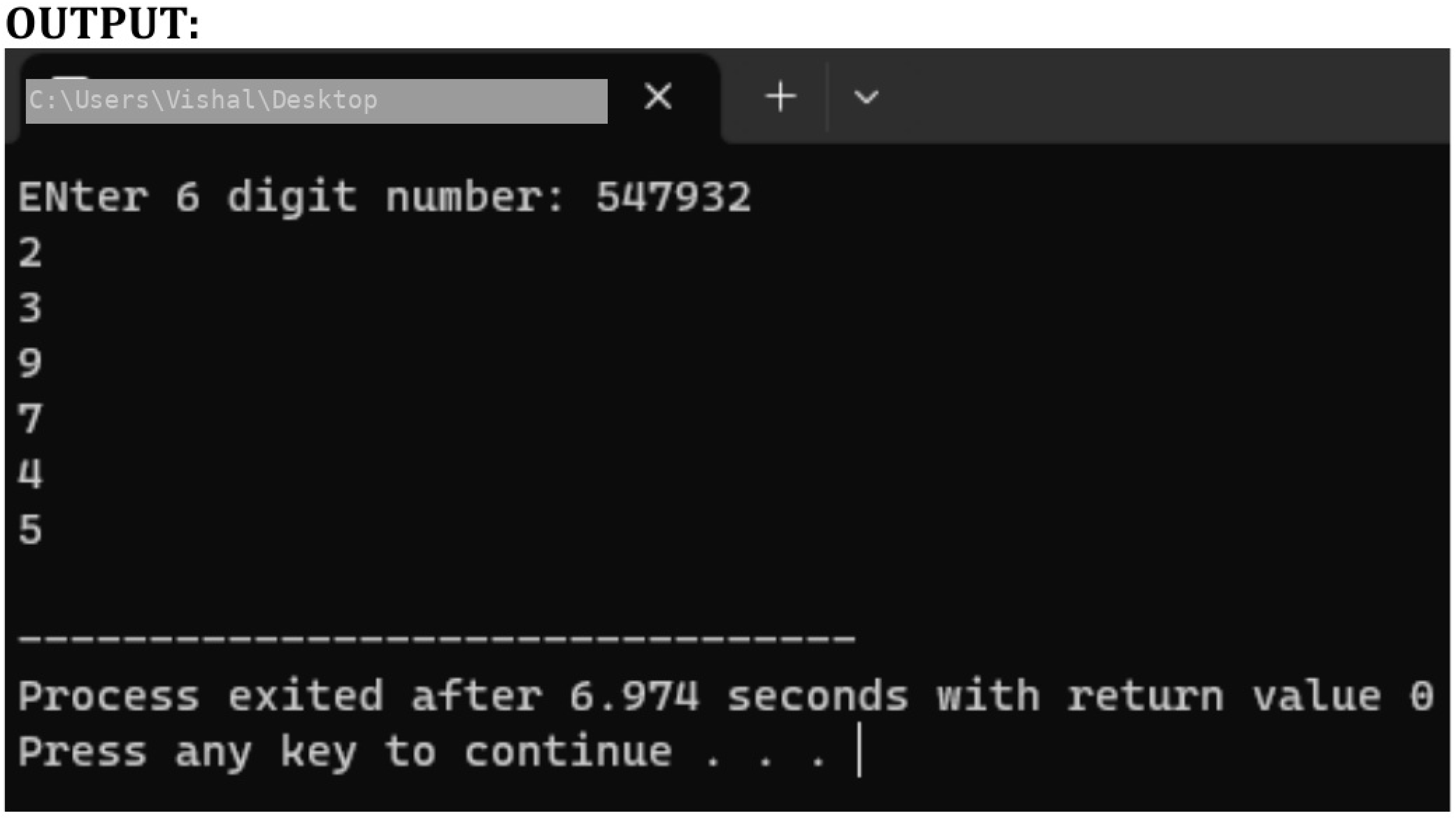
Write a program that inputs a six-digit integer, separates the integer into its individual digits and prints the digits vertically using modulus operator. For example, if the user types 953647, the program should print:

7
4
6
3
5
9

CODE:

```
#include <iostream>
using namespace std;
int main(){
	int num;
	cout<<"ENter 6 digit number: ";
	cin>>num;
	cout<<num%10<<endl;
	cout<<num/10%10<<endl;
	cout<<num/100%10<<endl;
	cout<<num/1000%10<<endl;
	cout<<num/10000%10<<endl;
	cout<<num/100000<<endl;
return 0;
}
```

OUTPUT:



QUESTION 3:

Write a program to convert Fahrenheit into Celsius.

$$C = \frac{5}{9}\,(F - 32)$$

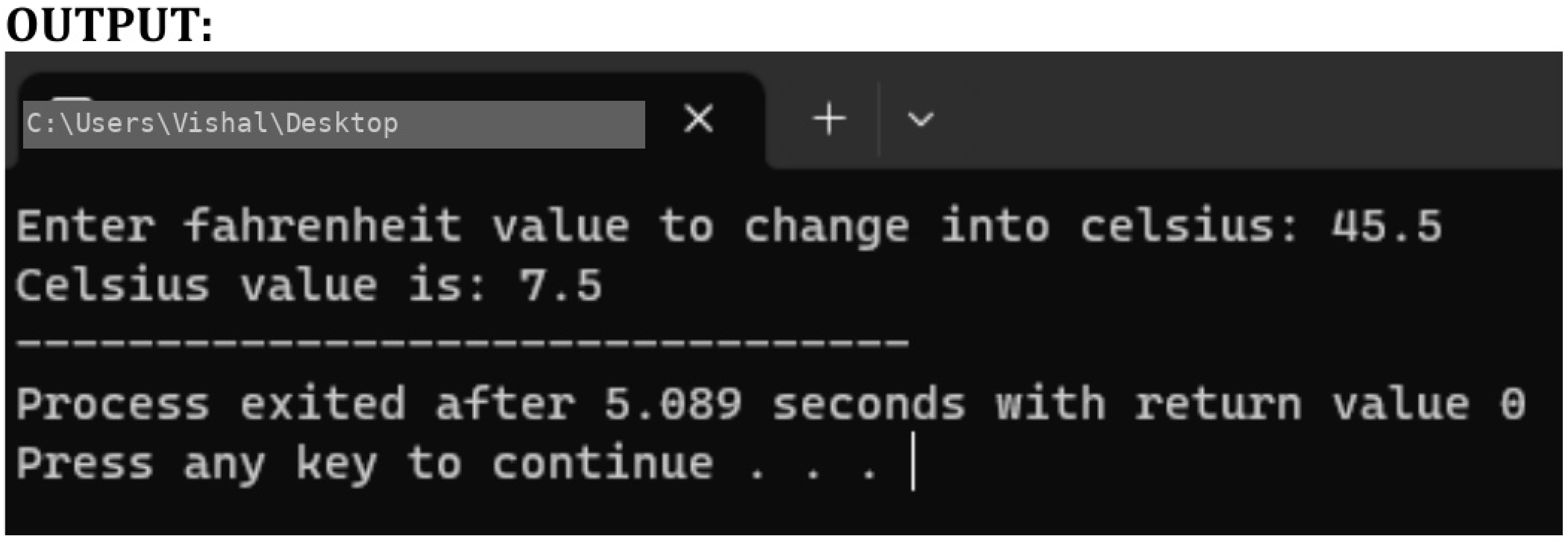
CODE:

```
#include <iostream>
using namespace std;
int main(){

double f,c;
	cout<<"Enter fahrenheit value to change into celsius: ";
	cin>>f;
c=(double)5/9*(f-32);
	cout<<"Celsius value is: "<<c;

return 0;
}
```

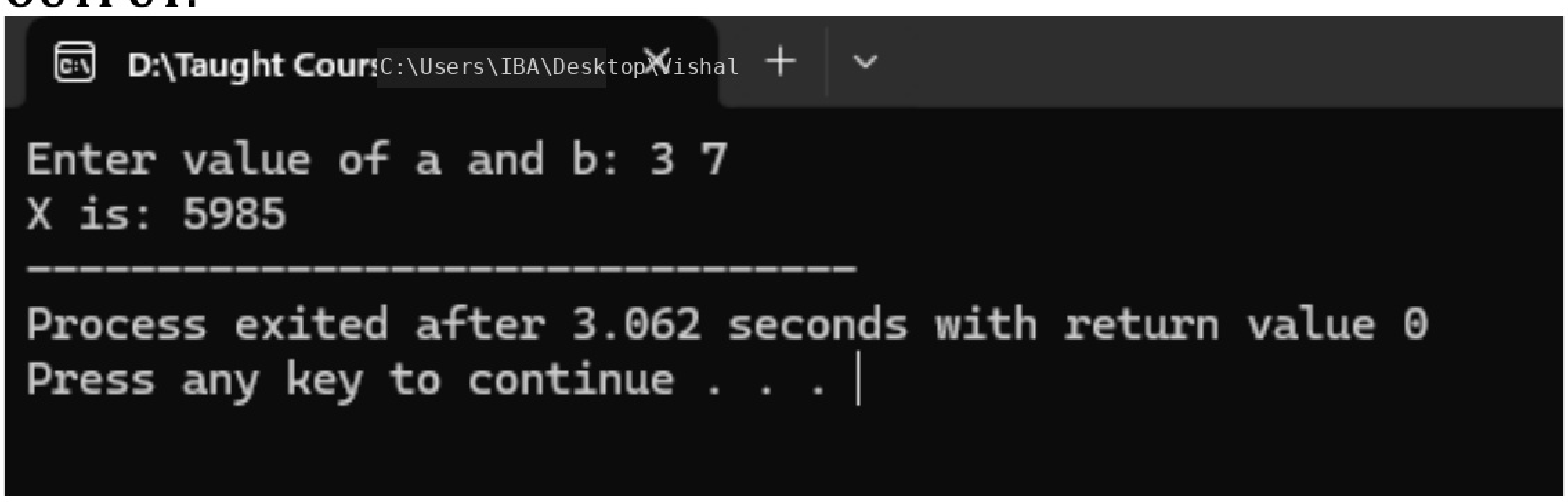
OUTPUT:



QUESTION 4:
Write a program that finds the value of X by using given formula. Take value of ‘a’ and ‘b’ from user.

$$X = 2a^2b\,(a + b^2) - 3a^3b$$

CODE:
```
#include<iostream>
using namespace std;
int main(){
	float a,b,X;
	cout<<"Enter value of a and b: ";
	cin>>a>>b;
	X=(2*a*a*b)*(a+b*b)-(3*a*a*a*b);
	cout<<"X is: "<<X;
return 0;
}
```

OUTPUT:


QUESTION 5:

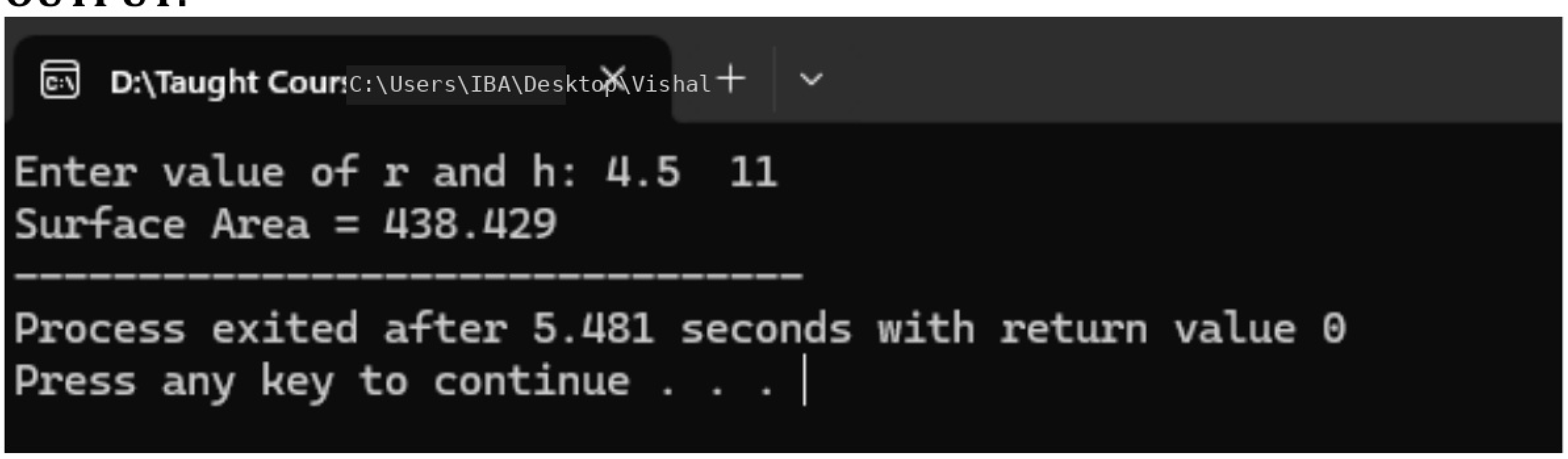
Create a program to calculate and display the surface of the cylinder given the radius (r) and height (h) using the formula:

$$Surface\_Area = 2\pi r^2 + 2\pi rh$$

CODE:

```
#include<iostream>
using namespace std;
int main(){
	float Surface_Area, r, h, pi=22.0/7;
	cout<<"Enter value of r and h: ";
	cin>>r>>h;
	Surface_Area=2*pi*r*r + 2*pi*r*h;
	cout<<"Surface Area = "<<Surface_Area;
return 0;
}
```

OUTPUT:



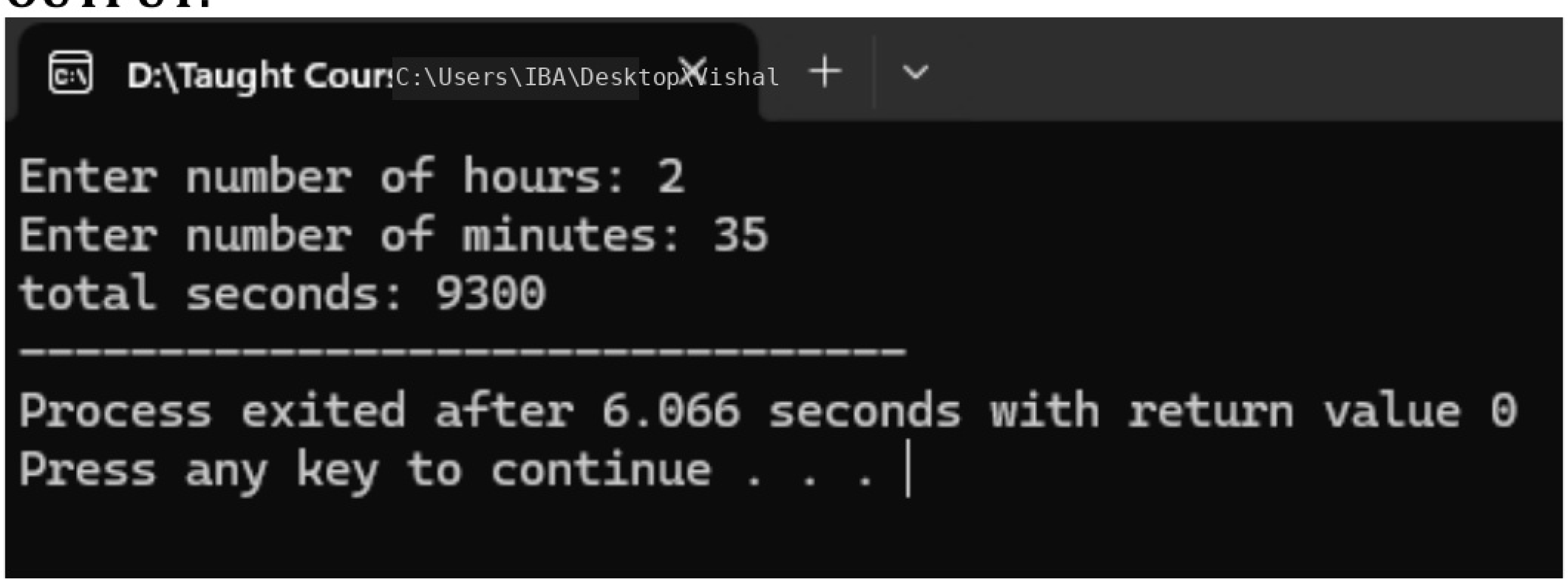
QUESTION 6:
Write a program that takes two values for hours and minutes and then calculates total seconds

CODE:

```
#include<iostream>
using namespace std;
int main(){
	int hour,mint,sec;
	cout<<"Enter number of hours: ";
	cin>>hour;
	cout<<"Enter number of minutes: ";
	cin>>mint;
	sec=(mint*60)+(hour*60*60);
	cout<<"total seconds: "<<sec;
return 0;

}
```

OUTPUT:



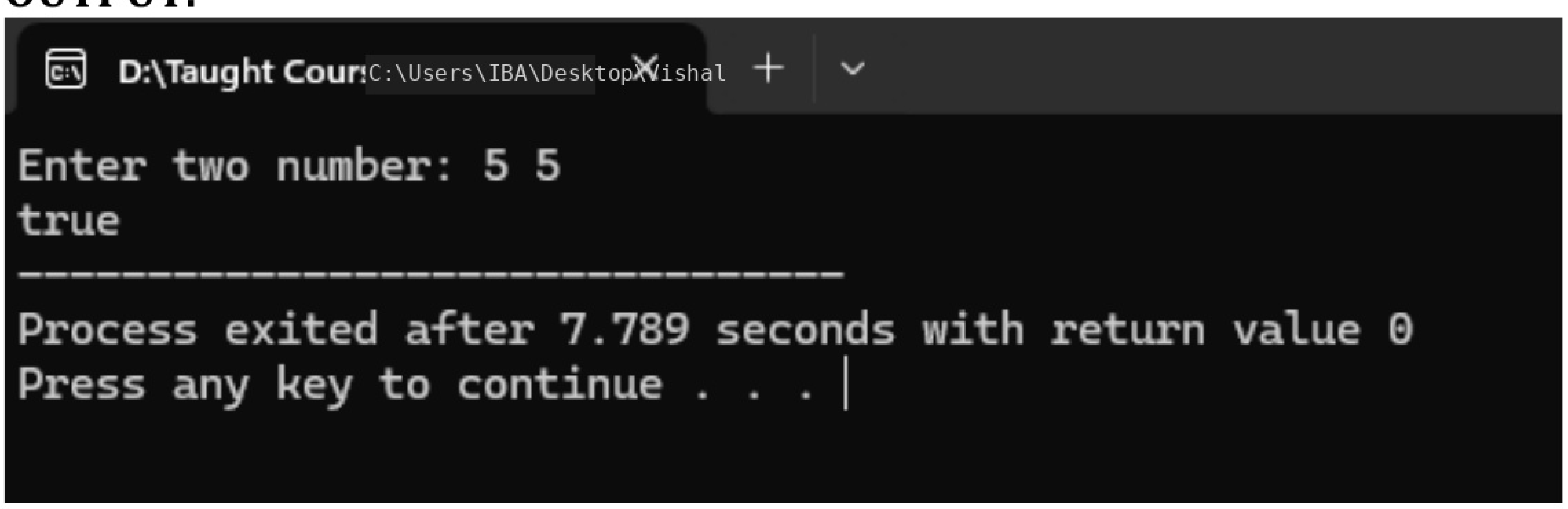
```
D:\Taught Cour: C:\Users\IBA\Desktop\Vishal
Enter number of hours: 2
Enter number of minutes: 35
total seconds: 9300
--------------------------------
Process exited after 6.066 seconds with return value 0
Press any key to continue . . .
```

QUESTION 7:
Write a program that takes two numbers from user and checks if the two numbers are equal or not by returning true or false respectively.

CODE:

```
#include <iostream>
using namespace std;
int main(){
	int a,b;
	bool result;
	cout<<"Enter two number: ";
	cin>>a>>b;
result=(a==b);
	cout<<boolalpha<<result;
return 0;
}
```

OUTPUT:

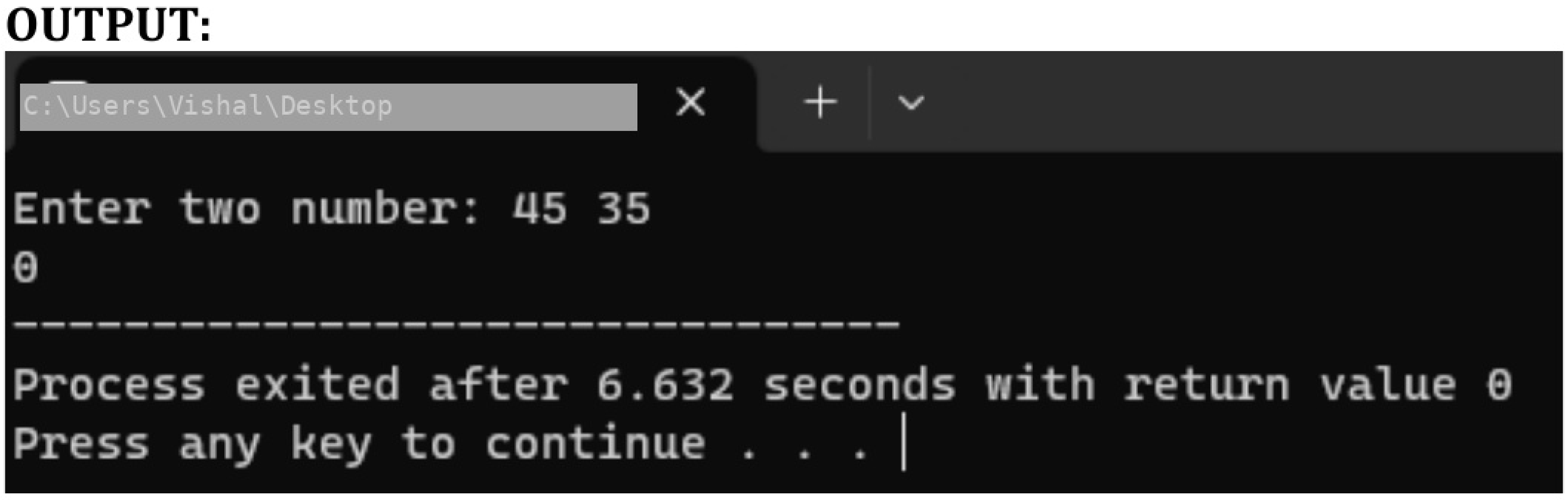


QUESTION 8:
Write a program to enter the values of two variables 'a' and 'b' from user and then check if both the conditions 'a < 50' and 'a < b' are true.

CODE:

```
#include <iostream>
using namespace std;
int main(){
        int a,b;
        bool result;
        cout<<"Enter two number: ";
        cin>>a>>b;
result= (a<50 && a<b);
        cout<<result;
return 0;
}
```

OUTPUT:

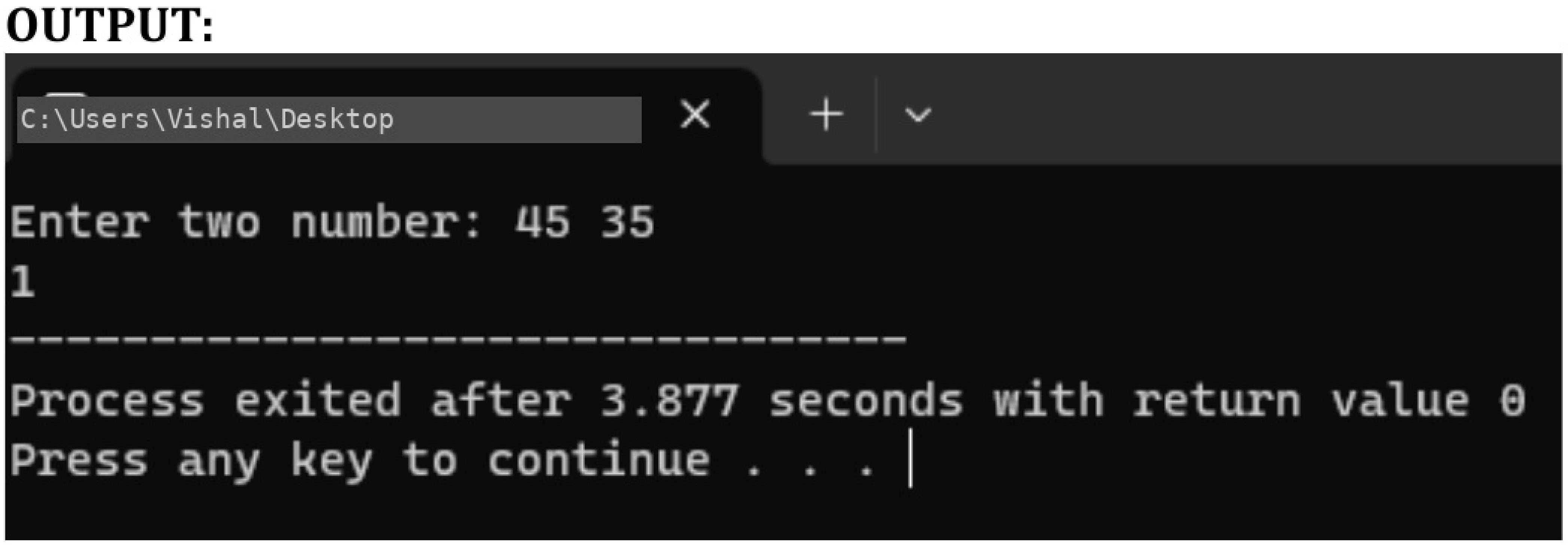


QUESTION 9:
Write a program to enter the values of two variables 'a' and 'b' from user and then check if atleast one of the conditions 'a < 50' and 'a < b' is true.

CODE:

```
#include <iostream>
using namespace std;
int main(){
        int a,b;
        bool result;
        cout<<"Enter two number: ";
        cin>>a>>b;
result= (a<50 || a<b);
        cout<<result;
return 0;
}
```

OUTPUT:



QUESTION 10:
Implement a program that calculates the square of a number entered by the user (using assignment operator).

CODE:

```
#include <iostream>
using namespace std;
int main(){
	int num;
	cout<<"Enter any number: ";
	cin>>num;
	cout<<"Square of "<<num<<" is: ";
num*=num;
	cout<<num;
return 0;
}
```

OUTPUT: